Exercise 6: Multivariate Optimization 1

Optimization in Machine Learning

> **Reproducibility**
>
> Produced with Python 3.10 and R 4.4.1. To reproduce, install the pinned package versions below.
>
> **Python**
>
> ```
> pip install numpy==2.0.2 matplotlib==3.10.8
> ```
>
> **R**
>
> ```
> install.packages("remotes")
> remotes::install_version("ggplot2", "3.5.1")
> remotes::install_version("gridExtra", "2.3")
> ```

Setup

You are given the following data situation:

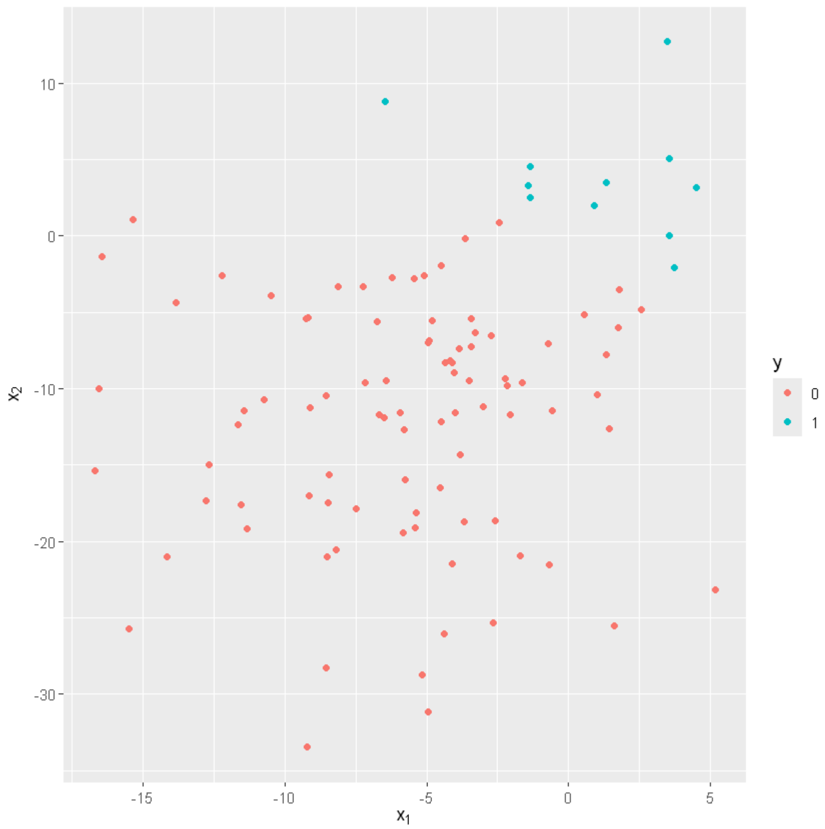
R

```
library(ggplot2)
library(gridExtra)

set.seed(314)
n <- 100
X <- cbind(rnorm(n, -5, 5), rnorm(n, -10, 10))
z <- 2 * X[, 1] + 3 * X[, 2]
```

```
pr <- 1 / (1 + exp(-z))
y <- as.integer(pr > 0.5)
df <- data.frame(X = X, y = y)

ggplot(df) +
  geom_point(aes(x = X.1, y = X.2, color = as.factor(y))) +
  xlab(expression(x[1])) +
  ylab(expression(x[2])) +
  labs(colour = "y")
```



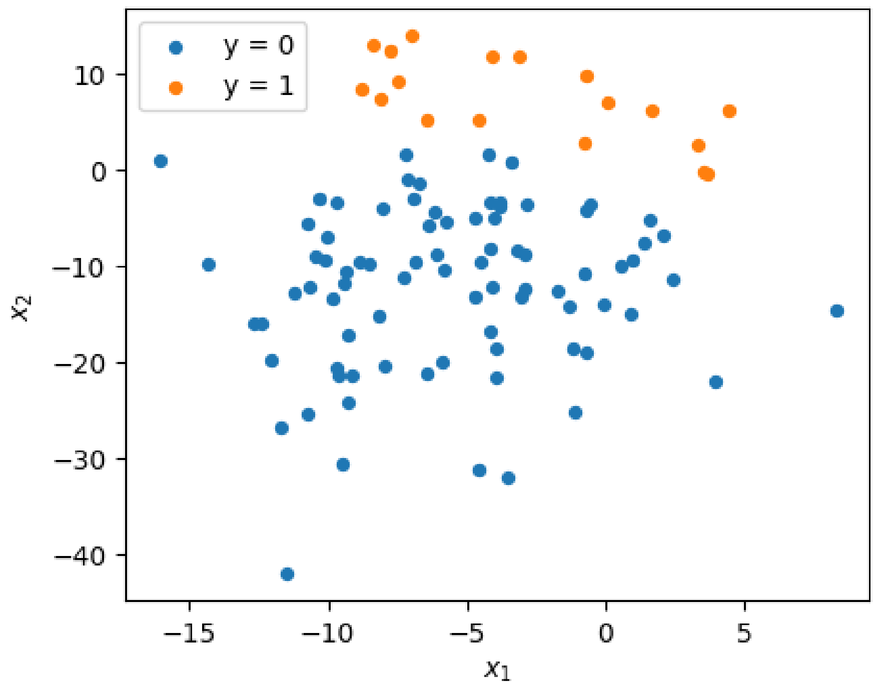
Python

```
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(314)
```

```
n = 100
X = np.column_stack([np.random.normal(-5, 5, n), np.random.normal(-10, 10, n)])
z = 2 * X[:, 0] + 3 * X[:, 1]
pr = 1 / (1 + np.exp(-z))
y = (pr > 0.5).astype(int)

fig, ax = plt.subplots(figsize=(5, 4))
for label in (0, 1):
    mask = y == label
    ax.scatter(X[mask, 0], X[mask, 1], label=f"y = {label}", s=18)
ax.set_xlabel(r"$x_1$"); ax.set_ylabel(r"$x_2$")
ax.legend()
plt.show()
```



ℹ Note

R vs. Python parity. R's `set.seed(314)` and Python's `np.random.seed(314)` use *different* RNGs. The data above (and consequently the GD trajectories below) differ between languages. Both implementations are correct; only the underlying data differs. Compare *algorithmic behavior* (divergence, convergence, backtracking adaptation) rather than expecting numerically identical θ values.

In the following we want to estimate a logistic regression *without intercept* via gradient descent.

Note: We use vanilla GD for educational purposes; in practice more sophisticated algorithms are typically preferred.

(a) Complete separation makes the empirical risk unbounded below

The data above is **completely separable**: the two classes can be perfectly classified with a linear classifier. Show that in this situation, if $\tilde{\theta}$ perfectly classifies the data then

$$\mathcal{R}_{\text{emp}}(\tilde{\theta}) > \mathcal{R}_{\text{emp}}(\alpha\tilde{\theta}) \quad \text{for all } \alpha > 1.$$

Implication: the per-observation log-loss can be driven arbitrarily low by scaling θ, so the empirical risk has no finite minimum.

(b) Visualize the empirical risk

Visualize $\mathcal{R}_{\text{emp}}$ on $[-1, 4] \times [-1, 4]$.

(c) Gradient of the empirical risk

Find $\nabla_\theta \mathcal{R}_{\text{emp}}$ for arbitrary θ.

(d) Vanilla gradient descent

Solve the logistic regression via gradient descent. Use step size $\alpha = 0.01$, starting point $\theta^{[0]} = (0, 0)^\top$, and train for 500 steps. Repeat with $\alpha = 0.02$. Explain what you observe.

Hint: recall (a).

(e) Add L2 regularization

Repeat (d), but minimize $\lambda\|\theta\|^2 + \mathcal{R}_{\text{emp}}(\theta)$ with $\lambda = 1$. What changes?

(f) Visualize the regularized empirical risk

Visualize the regularized empirical risk $\lambda\|\theta\|^2 + \mathcal{R}_{\text{emp}}(\theta)$ on $[-1, 4] \times [-1, 4]$ with $\lambda = 1$.

(g) Gradient descent with backtracking line search

Repeat (e) but with **backtracking line search**: at each step, propose $\theta_{\text{prop}} = \theta - \alpha \nabla f(\theta)$ and accept it if it satisfies the Armijo condition

$$f(\theta_{\text{prop}}) \leq f(\theta) - \gamma\, \alpha\, \|\nabla f(\theta)\|^2,$$

otherwise shrink $\alpha \leftarrow \tau\alpha$ and try again. Use $\gamma = 0.9$, $\tau = 0.5$.

Note: the trace below visualizes from iteration $t = 1$ (after one backtracking step), not from the starting point $\theta^{[0]} = (0, 0)^\top$.